



# Performances of Machine Learning in Detecting Glaucoma Using Fundus and Retinal Optical Coherence Tomography Images: A Meta-Analysis

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- **PURPOSE:** To evaluate the performance of machine learning (ML) in detecting glaucoma using fundus and retinal optical coherence tomography (OCT) images.
- **DESIGN:** Meta-analysis.
- **METHODS:** PubMed and EMBASE were searched on August 11, 2021. A bivariate random-effects model was used to pool ML's diagnostic sensitivity, specificity, and area under the curve (AUC). Subgroup analyses were performed based on ML classifier categories and dataset types.
- **RESULTS:** One hundred and five studies (3.3%) were retrieved. Seventy-three (69.5%), 30 (28.6%), and 2 (1.9%) studies tested ML using fundus, OCT, and both image types, respectively. Total testing data numbers were 197,174 for fundus and 16,039 for OCT. Overall, ML showed excellent performances for both fundus (pooled sensitivity = 0.92 [95% CI, 0.91-0.93]; specificity = 0.93 [95% CI, 0.91-0.94]; and AUC = 0.97 [95% CI, 0.95-0.98]) and OCT (pooled sensitivity = 0.90 [95% CI, 0.86-0.92]; specificity = 0.91 [95% CI, 0.89-0.92]; and AUC = 0.96 [95% CI, 0.93-0.97]). ML performed similarly using all data and external data for fundus and the external test result of OCT was less robust (AUC = 0.87). When comparing different classifier categories, although support vector machine showed the highest performance (pooled sensitivity, specificity, and AUC ranges, 0.92-0.96, 0.95-0.97, and 0.96-0.99, respectively), results by neural network and others were still good (pooled sensitivity, speci-

ficity, and AUC ranges, 0.88-0.93, 0.90-0.93, 0.95-0.97, respectively). When analyzed based on dataset types, ML demonstrated consistent performances on clinical datasets (fundus AUC = 0.98 [95% CI, 0.97-0.99] and OCT AUC = 0.95 [95% CI, 0.93-0.97]).

- **CONCLUSIONS:** Performance of ML in detecting glaucoma compares favorably to that of experts and is promising for clinical application. Future prospective studies are needed to better evaluate its real-world utility. (Am J Ophthalmol 2022;237: 1–12. © 2021 Elsevier Inc. All rights reserved.)

**G**LAUCOMA, A LEADING CAUSE OF BLINDNESS characterized by an insidious onset, is often first diagnosed only after irreversible vision impairment has occurred. Therefore, the identification of signs of glaucoma through clinical examinations is important.<sup>1</sup> In addition to measurements of intraocular pressure and the visual field, fundus photography and optical coherence tomography (OCT) are 2 other diagnostic modalities used to detect and monitor glaucoma.<sup>1-3</sup>

With an increasing prevalence and irreversible loss of visual function,<sup>4,5</sup> an efficient, sensitive, and specific tool for accurate detection and screening of glaucoma is needed. Because glaucoma evaluation by clinicians can be time- and labor-consuming, skills-demanding, and prone to inconsistency or subjectivity,<sup>6,7</sup> the application of automated image analysis by machine learning (ML) has been investigated as a tool to enhance glaucoma detection.

ML uses existing data to train and build a model that automatically performs certain tasks on a new dataset, providing a less human-dependent approach for medical image interpretation.<sup>8</sup> Ophthalmology employed ML early to classify healthy and diseased images,<sup>9</sup> and excellent results have been achieved for several common ocular diseases.<sup>10</sup> In addition to conventional ML algorithms, such as support vector machine (SVM); random forest; and k-nearest neighbor; deep learning, a newer ML technique that uses the neural network (NN) has transformed automated image analysis.<sup>11-13</sup> One notable example is the superior accuracy of NN in detecting diabetic retinopathy compared with hu-

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